

# DUSTON MOORE, PhD

FORMAL METHODS RESEARCHER | ROCQ/COQ · OCAML · TRUSTWORTHY AI

Edmonton, Canada | Open to relocation to Paris with support

dhwcmoore@gmail.com | 780-938-3102 | github.com/dhwcmoore | dhwcmoore.github.io

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## RESEARCH PROFILE

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Formal methods researcher and systems architect developing machine-checked, executable assurance systems for AI and other consequential software. I work across theorem statements, Rocq/Coq proofs, OCaml extraction and implementation, Python verifiers, adversarial tests, traceability, and trusted-computing-boundary documentation.

My central research question is how an untrusted generator, learned model, or operational system can produce claims whose authority comes from an independent checker rather than from the component that proposed them. Current work covers proof-carrying evidence, authenticated audit manifests, learned observation regimes, structural admissibility, and local-to-global obstruction.

My PhD is in Philosophy. A prior career as a tenured Associate Professor and Director developed research leadership, scientific writing, peer review, curriculum design, and cross-unit programme execution. My formal-methods record is demonstrated in the public proof developments and executable artefacts below.

## CORE EXPERTISE

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**Theorem proving:** Rocq/Coq, constructive proofs, inductive semantics, formal specification, proof-term and dependency audits, `Print Assumptions`, `coqchk`

**Verified and executable systems:** OCaml, Coq extraction, Python, exact arithmetic, command-line tools, property, fixture, differential, and adversarial testing

**Assurance architecture:** proof-carrying certificates, independent checkers, canonical encodings, authenticated manifests, policy and context binding, replay, provenance, explicit trust boundaries

**AI and systems research:** learned observation regimes, audit warrants, model-evidence separation, distributed consistency, sensor-fusion assurance, runtime monitoring

**Research execution:** research roadmaps, theorem ladders, verification plans, assumption registers, non-claims, reproducible builds, technical and stakeholder communication

**Mathematical methods:** finite linear algebra, exact rational arithmetic, quotient and factorisation analysis, obstruction methods, finite discrete and local-to-global structures

## SELECTED FORMAL METHODS RESEARCH

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### Proof-Carrying Fibre Witnesses for Learned Observation Regimes

*Creator and Lead Researcher | Rocq/Coq 8.18, OCaml | Active research, 2026*

Building a proof-carrying audit system for checking whether a learned observation interface has erased a distinction required by a declared target predicate.

- Proved the kernel result that a checked fibre witness refutes target-relative fibre constancy under an explicit policy.
- Developed typed campaign validation and capture/replay orchestration with first-failure ordering, fuel accounting, canonical manifest parsing, trust-anchor checks, and committed policy and context binding.
- Current source-only review build contains 14 kernel-checked modules, 71 closed assumption checks, and seven OCaml harnesses. Phase 1 remains open; full pipeline completion and verified cryptographic primitives are not claimed.

### DO-333 Rocq Assurance Case Study

*Creator | Rocq/Coq 8.18, Python, Make | Public release v1.0.0*

Completed an independent reconstruction of selected NASA/Rockwell Collins Flight Guidance System requirements arguments, organised around DO-178C/DO-333 evidence objectives without claiming certification credit.

- Reproduced three source-reported requirement defects as machine-checked refutations, then proved the corrected requirements over their explicit domains.
- Constructively proved the requirement context inhabited and tested each of eleven high-level requirements by removal.
- Published a 223-result inventory, proof-footprint audits, traceability checks, assumptions, discrepancies, non-claims, tool-trust boundary, and completion report; `coqchk` and repository boundary checks run on every verification build.

github.com/dhwcmoore/do333-rocq-assurance-case-study

## Proof-Carrying Audit Warrants

*Creator | Rocq/Coq 8.18, extracted OCaml | Public v0.1.0 candidate*

Developed a claim-relative warrant layer for dynamic AI audit evidence, separating a mitigation's effect from the continuing authority of the evidence offered for it.

- Formalised warranted, defeated, underdetermined, and inadmissible verdicts, with explicit obligations and witnesses.
- Proved soundness and completeness between a declarative semantics and executable classifier, including preservation and defeat under evidence extension.
- Built an extracted checker and CLI; 14 public fixtures across three source-gated AI-audit cases agree with the formal decisions. `coqchk` clean, with no admitted obligations or project axioms.

[github.com/dhwcmoore/proof-carrying-audit-warrants](https://github.com/dhwcmoore/proof-carrying-audit-warrants)

## Regional Obstruction Calculus

*Creator | Rocq 9.x, OCaml, Python | Public research development*

Developed exact rational and machine-checked methods for detecting and auditing local-to-global coherence failure in finite regional systems.

- Separates repairability, obstruction, refinement persistence, descent, reflection, and presentation-relative verdict invariance rather than treating them as one claim.
- Produces constructive repairs or separating witnesses and proof-carrying certificates checked independently with exact arithmetic.
- The active 37-module Rocq chain is checked with `coqc` and `coqchk`; the repository states stronger presentation-invariance claims that remain open.

[github.com/dhwcmoore/regional-obstruction-calculus](https://github.com/dhwcmoore/regional-obstruction-calculus)

## PROFESSIONAL EXPERIENCE

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### Independent Formal Methods Researcher and Systems Architect

*Edmonton, Canada | 2023–present*

Define and execute independent research programmes in formal verification and assurance for AI, distributed evidence, and automated systems.

- Carry projects from governing claim and threat model through formalisation, implementation, extraction, test harnesses, verification records, and public release.
- Design proof-carrying formats and independent checking boundaries that preserve provenance, assumptions, unresolved obligations, and failure witnesses.
- Write research manuscripts and extensive technical documentation for expert and cross-disciplinary readers.

### Education Developer

*University of Alberta | 2015–2016*

Designed programmes and assessments, led workshops and peer review, supported educational technology, and worked across faculties.

### Associate Professor of Philosophy / Director of General Education Assessment

*Purdue University Fort Wayne | 2002–2013*

- Published and taught as a tenured researcher; reviewed scholarly work and participated in university governance.
- Directed institution-wide assessment involving evidence standards, programme requirements, reporting, workshops, and collaboration across independent academic units.

## EDUCATION

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### PhD, Philosophy

*KU Leuven, Belgium*

Doctoral and academic research in argument, evidence, process ontology, and conceptual structure. Subsequent independent work moved these concerns into formal specification, machine-checked proof, and executable verification.

## SELECTED WRITING

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**Admissibility Defines Structural Limits on Optimisation in Artificial Intelligence.** Accepted for publication in *Discover Artificial Intelligence* (Springer Nature). Read the accepted manuscript (PDF).

Earlier publications include “Marcuse and Eternal Objects”, *Process Studies*, and “Propositions in Corporations”, *Applied Process Thought II* (2008).